

Deep Space Cargo Recovery

A 3D space simulation and precision recovery game developed with Three.js and Vite. Pilot an exploration probe, manage solar battery levels, navigate orbital hazards, and recover precious energy crystals to transport them safely back to the mothership cargo bay.

 [Play the game directly here!](#)

LINK DEMO: https://sapienzainteractivegraphicscourse.github.io/final-project-space_groovers/

Features & Highlights

Game Modes

- **Testing the Probe (Standard):** Free exploration with standard battery drain, unlimited time, and clear flight corridors.
 - **Time Attack:** 2-minute mission countdown with static orbital satellite debris functioning as destructive hazards.
 - **Spinning out of Space:** Hardcore mode featuring 8 actively orbiting satellites, floating micro-asteroid fields, a 90-second base timer, and a +20s Time Bonus per delivered crystal.
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Controls Reference

Input Key / Action	Function
W / S	Forward / Backward Thrust
A / D	Roll Left / Right
Left / Right Arrows	Yaw Left / Right
Up / Down Arrows (or Q / E)	Pitch Up / Down
R	Extend / Retract Robotic Arm
G	Grab / Release Crystal
P	Deploy / Fold Solar Panels (Recharge)
C	Toggle Cockpit (First-Person) / Third-Person Camera
Left Click + Drag	Orbit Camera View
Mouse Wheel	Zoom In / Out

Tech Stack

- **Core Engine:** Three.js (WebGL 3D Rendering)

- **Animation & Interpolation:** [@tweenjs/tween.js](#)
 - **Build Tool & Dev Server:** Vite
 - **Debug Interface:** [lil-gui](#)
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Getting Started

Prerequisites

Ensure you have **Node.js** (v16 or newer) installed on your system.

Installation & Run

If you want to run the game locally, follow these steps:

1. Clone the repository and navigate into the project directory:

```
git clone [https://github.com/SapienzaInteractiveGraphicsCourse/final-project-space_groovers]
cd final-project-space_groovers
```

2. Install all required dependencies:

```
npm install
```

3. Launch the local development server:

```
npm run dev
```

4. Open the displayed local address (usually <http://localhost:5173>) in any WebGL-compatible modern browser.

Project Structure

```
├── public/
│   ├── background-music.mp3
│   └── textures/
├── src/
│   ├── components/
│   │   ├── CargoShip.js
│   │   ├── DataGalaxy.js
│   │   └── Lights.js
```

```
|
| | └─ SpaceProbe.js
| └─ ui/
|   └─ UserInterface.js
|   └─ main.js
└─ index.html
└─ package.json
└─ README.md
```